

Sachit Madaan

(317)-288-7703 | sachhg21@gmail.com | GitHub: github.com/sachhg | LinkedIn: linkedin.com/in/madaan-sachit/

EDUCATION

UC Santa Barbara

BS, Computer Science (Honors): GPA: 3.9/4.0

Santa Barbara, CA

Sep 2024 - Dec 2027

EXPERIENCE

American Express (Software Engineer Intern) | *LangGraph, Python, LightGBM* (May 2026 – Aug 2026)

- Built and deployed an ML-driven autoscaling service for the internal DBaaS platform managing 500+ on-premises database clusters, forecasting per-cluster CPU, memory, IOPS, connection, and throughput demand and executing right-sizing actions automatically; replaced a manual ticket-and-approval process, cutting manual approvals 60% and fleet compute cost 8% at flat latency.
- Trained LightGBM forecasters on historical cluster telemetry to predict utilization 7 days ahead per resource dimension (12% MAPE), driving scale decisions across compute, memory, and storage rather than reacting to threshold breaches after saturation.

Conmitto (Software Engineer Intern) | *ASP.NET, C#, SQL, Azure, GitHub Actions* (Jun 2025 – Present)

- Designed and shipped a self-serve integration platform end to end, letting customers connect their own WMS, TMS, ERP, and IoT systems and sync live inventory, order, and shipment data into a multi-tenant .NET 8 dock scheduling product; the platform anchored the product story in a \$6M seed at a \$24M valuation and has synced 50K+ records across 15 production connectors.
- Engineering with ASP.NET Core around a canonical EF Core schema with per-tenant credential storage (OAuth), configurable field mapping, scheduled polling and push-webhook ingestion, per-connector sync history and error surfacing, and a generic REST adapter that onboards new sources through config rather than code.
- Hardened ingestion against partner-API failure using hosted-service background workers with idempotency keys, exponential-backoff retry, and dead-letter replay; separately cut p95 page load 12s → 3s via EF Core query tuning and a tiered IMemoryCache layer (60% DB queries).

Turing (Applied AI Intern) | *LangChain, LangGraph RAG* (Feb 2026 – May 2026)

- Built a LangGraph agent that triages inbound support calls in real-time from 10K+ property managers daily, running RAG retrieval over a 10K-record Chroma index into LLM extraction and classification stages; achieved 98.8% category accuracy and 95.4% auto-resolution rate.
- Optimized calling pipeline to 6.7s end-to-end against a P95 budget of 15s, isolating LLM stages as 95% of wall-clock versus sub-1.2s retrieval.

Raytheon (Software Consultant) | *C++23, CMake, FFTW3, OpenBLAS, LightGBM* (Jan 2026 – May 2026)

- Built a C++23 inference engine that classifies RF signals by modulation type on embedded RISC-V targets, exposed over a TCP socket with CMake-selectable OpenBLAS, SIMD, and scalar compute backends.
- Implemented the tensor and expression-template system that fuses element-wise ops into single loops, plus the convolution, normalization, pooling, and dense-layer primitives behind a type-safe FFTW3 wrapper for fast Fourier transforms.

PricewaterhouseCoopers (PwC) (Software Consultant) | *Flask, Next.js, sklearn* (Mar 2025 – May 2025)

- Built anomaly detection and financial forecasting models on millions of real and synthetic bank transactions.
- Developed anomaly detection using an Isolation Forest model and a deep autoencoder, cutting false positives by 25%.

TECHNICAL SKILLS

Languages: C++ (17/23), Python, C#, Java, SQL, JavaScript/TypeScript, HTML/CSS

Frameworks: ASP.NET Core, .NET 8, Entity Framework Core, FastAPI, Flask, Next.js, React, Node.js, TailwindCSS

Infrastructure: Azure, AWS, Docker, CI/CD, Linux, CMake, REST APIs, PostgreSQL, MongoDB, Supabase, Redis

ML & Data: LangGraph, LangChain, RAG, LightGBM, PyTorch, scikit-learn, Chroma, vector search, Pandas, NumPy, FFTW3, OpenBLAS

PROJECTS

pqc-scan | *Python, tree-sitter, SARIF, CycloneDX, GitHub Actions* (Jun 2026)

- Built a CLI tool and GitHub Action that scans codebases for cryptography vulnerable to quantum attack (RSA, ECDSA, Diffie-Hellman), flagging call sites and dependencies and failing CI when non-quantum-safe primitives are introduced; published on PyPI.
- Implemented AST-based static analysis over tree-sitter grammars for Python, JavaScript, Java, and Go, resolving import origins and call shapes, with separate config and dependency-manifest scanners; emits SARIF 2.1.0 for GitHub code scanning and a CycloneDX CBOM.